Week 10 Handout – Dynamic Programming

1- A political campaign is entering its final stage, and polls indicate a very close election. One of the candidates has enough funds left to purchase TV time for a total of 5 prime-time commercials on TV stations in 4 different areas. Based on polling data, an estimate has been made of the additional votes that can be won in the different broadcasting areas, depending on the number of commercials run. These estimates are given in the following tables in thousands of votes:

	Area			
# of Commercials	**1**	**2**	**3**	**4**
0	0	0	0	0
1	4	6	5	3
2	7	8	9	7
3	9	10	11	12
4	12	11	10	14
5	15	12	9	16

Use dynamic programming to determine how to distribute the five commercials across the four areas to maximize the estimated number of votes.

Let:

- $N = 4$ be the number of areas (stages of the DP).
- $W = 5$ be the total number of commercials available (the state variable).
- $v_i(x_i)$ be the estimated additional votes (in thousands) from area i given x_i commercials.
- $f_i(s)$ be the maximum number of votes that can be won from areas $i = 1, \cdots, 4$ given s commercials are available.

The recursive relationship is defined as:

$$f_i(s) = \max_{0 \le x_i \le s} \{ v_i(x_i) + f_{i+1}(s - x_i) \}$$

The base case is $f_5(s) = 0$ for all s.

Stage: $n = 4$		
s_4	$f_4(s_4)$	x_4^*
0	0	0
1	3	1
2	7	2
3	12	3
4	14	4
5	16	5

Stage: $n = 3$								
s_3	$f_3(s_3)$						$f_3^*(s_3)$	x_3^*
	0	1	2	3	4	5		
0	0+0=0	-	-	-	-	-	0	0
1	0+3=3	5+0=5	-	-	-	-	5	1
2	0+7=7	5+3=8	9+0=9	-	-	-	9	2
3	0+12=12	5+7=12	9+3=12	11+0=11	-	-	12	0, 1, 2
4	0+14=14	5+12=17	9+7=16	11+3=14	10+0=10	-	17	1
5	0+16=16	5+14=19	9+12=21	11+7=18	10+3=13	9+0=9	21	2

Stage: $n = 2$								
s_2	$f_2(s_2)$						$f_2^*(s_2)$	x_2^*
	0	1	2	3	4	5		
0	0+0=0	-	-	-	-	-	0	0
1	0+5=5	6+0=6	-	-	-	-	6	1
2	0+9=9	6+5=11	8+0=8	-	-	-	11	1
3	0+12=12	6+9=15	8+5=13	10+0=10	-	-	15	1
4	0+17=17	6+12=18	8+9=17	10+5=15	11+0=11	-	18	1
5	0+21=21	6+17=23	8+12=20	10+9=19	11+5=16	12+0=12	23	1

Stage: $n = 1$								
s_1	$f_1(s_1)$						$f_1^*(s_1)$	x_1^*
	0	1	2	3	4	5		
0	0+23=23	-----	-----	-----	-----	-----	23	0
1	-----	4+18=22	-----	-----	-----	-----	22	1
2	-----	-----	7+15=22	-----	-----	-----	22	2
3	-----	-----	-----	9+11=20	-----	-----	20	3
4	-----	-----	-----	-----	12+6=18	-----	18	4
5	-----	-----	-----	-----	-----	15+0=15	15	5

Optimal Solution:

$$x_1^* = 0,\ \ x_2^* = 1,\ \ x_3^* = 1,\ \ x_4^* = 3$$